(1) What is Institutional Memory?

→ Institutional memory in the context of online experiments (like A/B testing) is the collective knowledge of all past experiments an organization has run — stored in a systematic and accessible way.

Instead of each team or employee running experiments in isolation, institutional memory means:

- You log and remember what was tested.
- You record the outcomes, metrics, insights, and even why something was a failure or success.
- Future experiments can use this history to avoid repeating mistakes or to start with better assumptions.

> Think of it as the "brain" of your experimentation program. The more it remembers, the smarter your decisions become.

(1) Real-world Analogy

Imagine a hospital tracking thousands of patients over the years:

- A doctor who tracks every patient's symptom, treatment, and result builds a medical knowledge base.
- So when a new patient arrives with similar symptoms, they don't start from scratch — they compare with past cases and make faster, more accurate treatment decisions.

Similarly, an experimentation platform with institutional memory lets data scientists learn from the past instead of making blind guesses every time.

(2) Math Behind it: Institutional Memory = Prior knowledge.

In Bayesian statistics:
- The prior comes from past knowledge (institutional memory)
- The likelihood is your current data
- The posterior (your belief after seeing current data) is more accurate.

If you've seen 100 past experiments with an average conversion rate of 20%, and your current test shows 23%, the institutional memory pulls your estimate slightly back toward 20% - reduces noise and over reaction.

(3) How it Connects to Other Topics.

Topic	How it's Related.
Meta-Analysis	Combines multiple past experiment to generate insights — institutional memory is the input to meta-analysis.
OEC (Overall Evaluation Criteria)	Past experiments help fine-tune what you define as success (e.g. bounce rate vs. long-term retention)
Goodhart's Law	Past experiments show where metrics were gamed, which helps evolve guardrails.
Experiment Culture	Promotes transparency and prevents teams from "repeating history" unknowingly.

(4) Institutional Memory Tools (when you're ready)

Some companies build this memory using:

- Internal experiment dashboards (e.g. Airbnb's ERF, Netflix's Metaflow)
- Auto-logging tools for experiment metadata
- Bayesian priors injected into stats engines

(2) Why is Institutional Memory Useful?

→ Institutional memory is useful because it allows organizations to:

- Avoid repeating failed experiments.
- Learn from past results to improve future decisions
- Speed up decision-making by reusing insight.
- Improve statistical estimates using past knowledge (via Bayesian priors)
- Optimize resource use - fewer experiments, more impact.

(1) Simple Explanation

Running online experiment without institutional memory is like:

> "Trying to solve a jigsaw puzzle where everyone throws away the finished pieces".

With institutional memory:

- You build a growing knowledge base of what works and what doesn't.
- You reuse patterns (e.g. certain UX changes always hurt retention).

- You can identify global truths across many experiments (meta-analysis).
- You gain more power to detect true effects by combining current and past data.

(2) Real-World Analogy

Imagine you're a doctor treating diseases,

- Without institutional memory: You treat each patient like a fresh mystery.
- With memory: You recall similar past cases, what worked, what failed, and how long it took.

This lets you diagnose faster, more accurately, and void harmful mistakes

Now, apply that to online experiments memory lets you make better product decisions, faster.

(3) Overlapping Topics

Topic	Why it's linked
Bayesian A/B testing	Core math behind why institutional memory improves inference
Meta-analysis	Uses institutional memory to find consistent patterns across experiments.
Goodhart's Law	Institutional memory can alert you if a metric was previously gamed
OEC refinement	Long-term history helps redefine better success criteria.
Experiment power	You can pre-load more confidence into new tests if similar ones succeeded often.